

```

import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats

# --- Step 1: Load and Explore the Dataset ---
print("\n--- Step 1: Loading and Exploring the Dataset ---")
# Load the dataset (assuming the file is available in the current environment)
df = pd.read_csv('/content/marketing_and_sales_data_evaluate_lr.csv')

print("Dataset Head:")
print(df.head())
print("\nDataset Info (before cleaning):")
df.info()

# --- Step 2: Handle Missing Values ---
print("\n--- Step 2: Checking for Missing Values & Initial Cleaning ---")
print("Missing values per column (before dropping):")
print(df.isnull().sum())

# Drop rows with any missing values to ensure a clean dataset for regression
df.dropna(inplace=True)
print("\nDataFrame Info (after dropping rows with missing values):")
df.info()

```

--- Step 1: Loading and Exploring the Dataset ---

Dataset Head:

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

Dataset Info (before cleaning):

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 4572 entries, 0 to 4571

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	TV	4562 non-null	float64
1	Radio	4568 non-null	float64
2	Social_Media	4566 non-null	float64
3	Sales	4566 non-null	float64

dtypes: float64(4)

memory usage: 143.0 KB

--- Step 2: Checking for Missing Values & Initial Cleaning ---

Missing values per column (before dropping):

TV	10
Radio	4
Social_Media	6
Sales	6

```
dtype: int64
```

```
DataFrame Info (after dropping rows with missing values):
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 4546 entries, 0 to 4571
```

```
Data columns (total 4 columns):
```

#	Column	Non-Null Count	Dtype
0	TV	4546 non-null	float64
1	Radio	4546 non-null	float64
2	Social_Media	4546 non-null	float64
3	Sales	4546 non-null	float64

```
dtypes: float64(4)
```

```
memory usage: 177.6 KB
```

```
# --- Step 2 (continued): Descriptive Statistics ---
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```
print("\nDescriptive Statistics (after cleaning):")
```

```
print(df.describe())
```

```
Descriptive Statistics (after cleaning):
```

	TV	Radio	Social_Media	Sales
count	4546.000000	4546.000000	4546.000000	4546.000000
mean	54.062912	18.157533	3.323473	192.413332
std	26.104942	9.663260	2.211254	93.019873
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.555355	1.530822	112.434612
50%	53.000000	17.859513	3.055565	188.963678
75%	77.000000	25.640603	4.804919	272.324236
max	100.000000	48.871161	13.981662	364.079751

```
# --- Step 3: Perform Exploratory Data Analysis (EDA) and Variable Selection ---
```

```
print("\n--- Step 3: Exploratory Data Analysis (EDA) and Variable Selection ---")
```

```
# Calculate correlation matrix
```

```
corr = df.corr(numeric_only=True)
```

```
print("\nCorrelation Matrix:")
```

```
print(corr)
```

```
# Identify marketing channels correlation with Sales
```

```
print("\nMarketing channels correlation with Sales:")
```

```
print(corr['Sales'][['TV', 'Radio', 'Social_Media']].sort_values(ascending=False))
```

```
--- Step 3: Exploratory Data Analysis (EDA) and Variable Selection ---
```

```
Correlation Matrix:
```

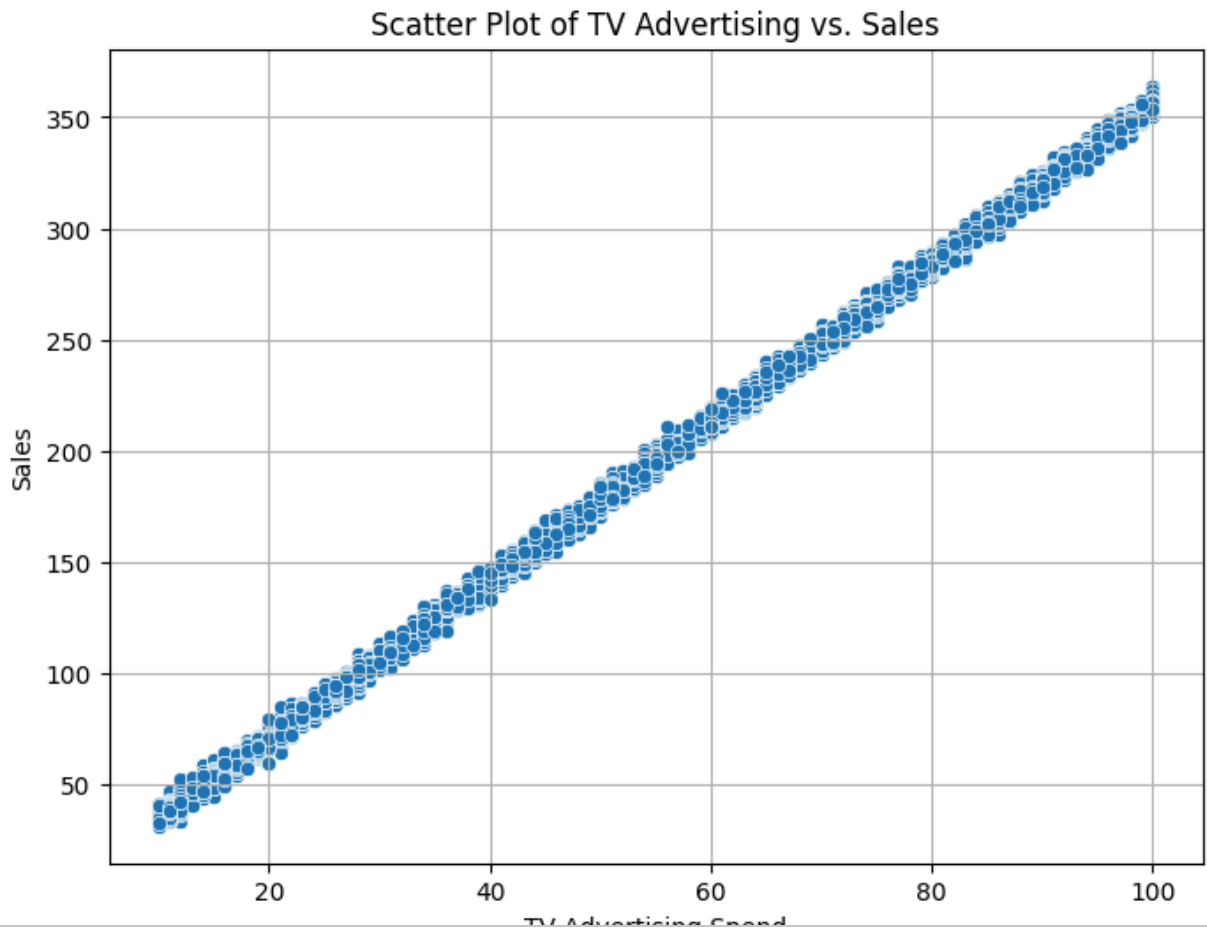
	TV	Radio	Social_Media	Sales
TV	1.000000	0.869158	0.527687	0.999497
Radio	0.869158	1.000000	0.606338	0.868638
Social_Media	0.527687	0.606338	1.000000	0.527446
Sales	0.999497	0.868638	0.527446	1.000000

```
Marketing channels correlation with Sales:
```

```
TV 0.999497
```

```
Radio          0.868638
Social_Media   0.527446
Name: Sales, dtype: float64
```

```
the relationship between TV and Sales (as it had the highest individual correlation)
figsize=(8, 6))
plt.scatter(x='TV', y='Sales', data=df)
# Scatter Plot of TV Advertising vs. Sales')
# TV Advertising Spend')
# Sales')
# e)
```



```
# --- Step 4: Build a Multiple Linear Regression Model using statsmodels ---
print("\n--- Step 4: Building the Multiple Linear Regression Model (TV, Radio,

# Define the independent variables (X) and dependent variable (y)
X = df[['TV', 'Radio', 'Social_Media']]
y = df['Sales']

# Add a constant to the independent variables for the intercept term
X = sm.add_constant(X)
```

```
--- Step 4: Building the Multiple Linear Regression Model (TV, Radio, Soc
```

```
# Create and fit the OLS model
model = sm.OLS(y, X).fit()

# Print the full OLS regression summary table
print("\nOLS Regression Results (Multiple Linear Regression):")
print(model.summary())
```

OLS Regression Results (Multiple Linear Regression):

OLS Regression Results

```
=====
Dep. Variable:          Sales    R-squared:
Model:                  OLS      Adj. R-squared:
Method:                Least Squares    F-statistic:          1.50
Date:                  Wed, 10 Jun 2026    Prob (F-statistic):
Time:                  15:48:42    Log-Likelihood:        -1
No. Observations:      4546    AIC:                    2.27
Df Residuals:          4542    BIC:                    2.27
Df Model:              3
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025
const	-0.1340	0.103	-1.303	0.193	-0.336
TV	3.5626	0.003	1051.118	0.000	3.556
Radio	-0.0040	0.010	-0.406	0.685	-0.023
Social_Media	0.0050	0.025	0.199	0.842	-0.044

```
=====
Omnibus:                0.056    Durbin-Watson:
Prob(Omnibus):           0.972    Jarque-Bera (JB):
Skew:                   -0.001    Prob(JB):
Kurtosis:               3.013    Cond. No.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is co

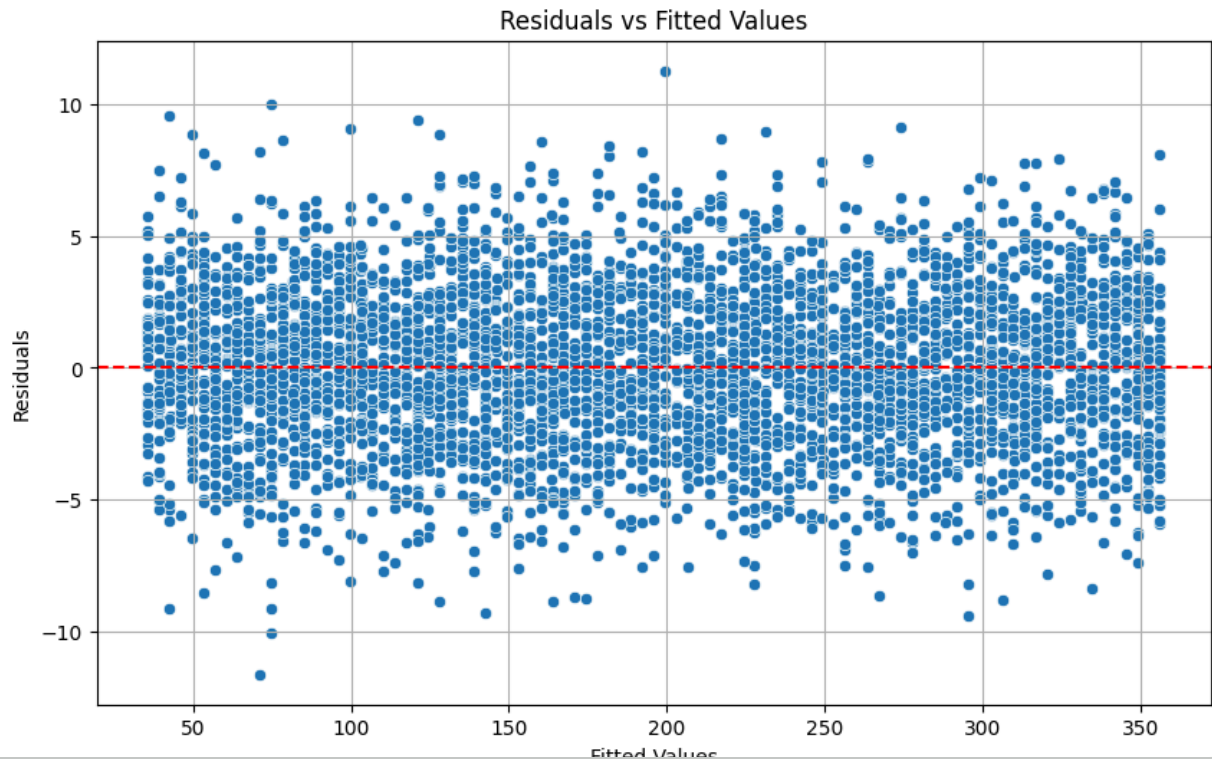
```
# --- Step 5: Create Diagnostic Plots to Test Assumptions ---
print("\n--- Step 5: Creating Diagnostic Plots for Assumption Checking ---")

# Get residuals and fitted values
predictions = model.predict(X)
residuals = model.resid
```

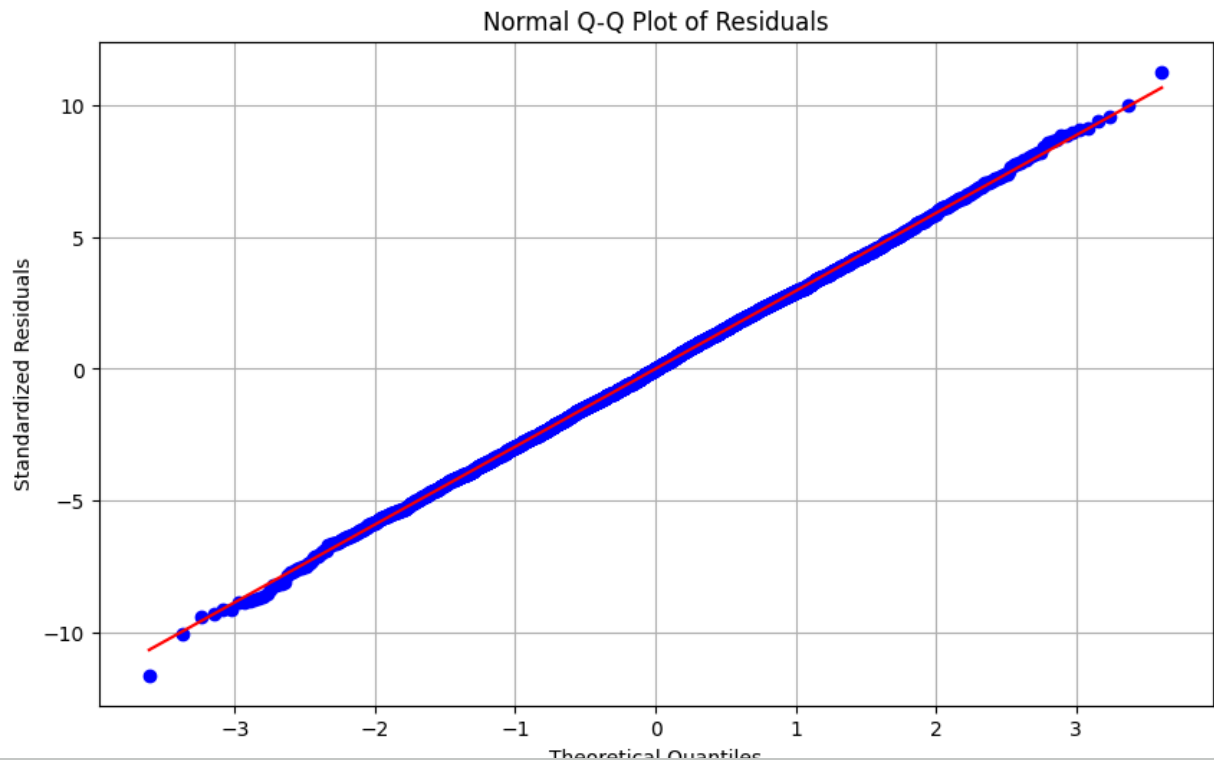
--- Step 5: Creating Diagnostic Plots for Assumption Checking ---

```
# 5.1 Residuals vs Fitted Plot (Linearity and Homoscedasticity)
plt.figure(figsize=(10, 6))
sns.scatterplot(x=predictions, y=residuals)
plt.axhline(0, color='red', linestyle='--')
plt.title('Residuals vs Fitted Values')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
```

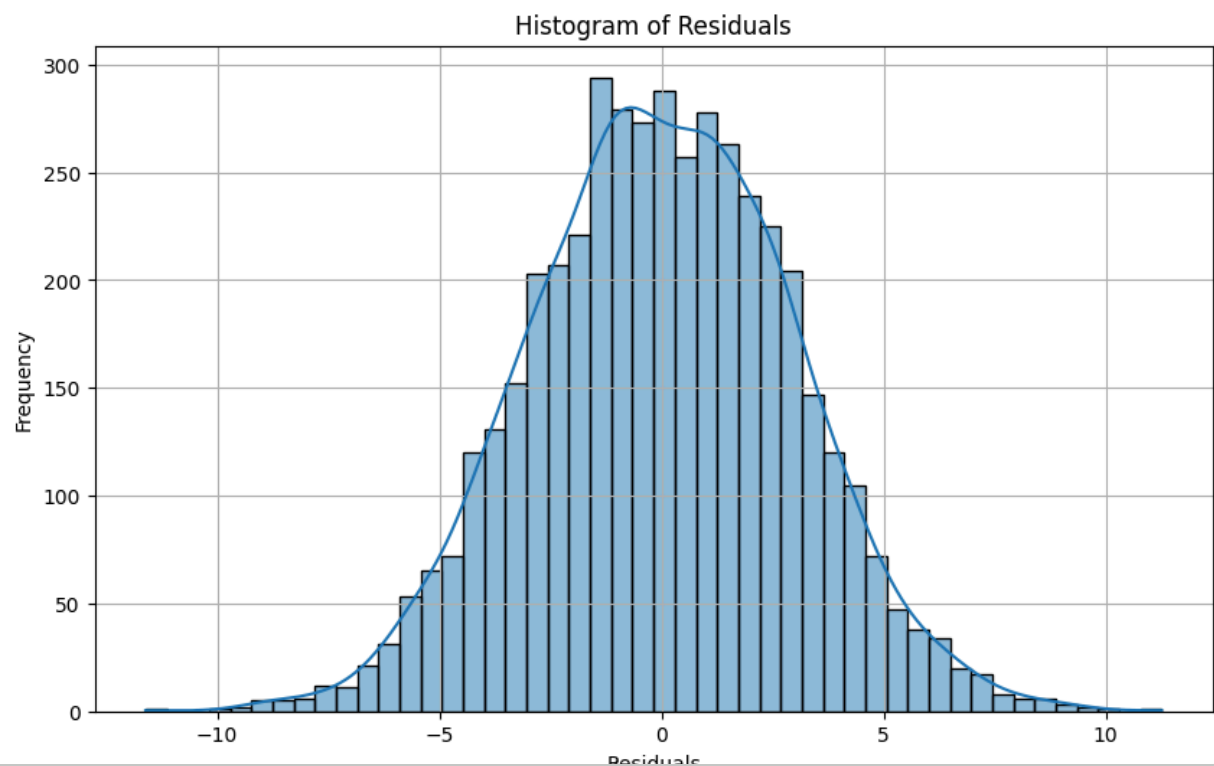
```
plt.grid(True)
plt.show()
```



```
# 5.2 Q-Q Plot of Residuals (Normality)
plt.figure(figsize=(10, 6))
stats.probplot(residuals, dist="norm", plot=plt)
plt.title('Normal Q-Q Plot of Residuals')
plt.xlabel('Theoretical Quantiles')
plt.ylabel('Standardized Residuals')
plt.grid(True)
plt.show()
```



```
# 5.3 Histogram of Residuals (Normality)
plt.figure(figsize=(10, 6))
sns.histplot(residuals, kde=True)
plt.title('Histogram of Residuals')
plt.xlabel('Residuals')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



Project Summary: Marketing ROI Analysis with Multiple Linear Regression

This project aimed to analyze a marketing dataset to understand the relationship between advertising spend on various channels (TV, Radio, Social Media) and Sales, ultimately providing an ROI-based recommendation for marketing budget allocation. A Multiple Linear Regression model was implemented using Python and `statsmodels`.

1. Data Loading and Cleaning:

- The `marketing_and_sales_data_evaluate_lr.csv` dataset was loaded.
- Initial data exploration revealed missing values across all marketing channels and Sales. These missing values were handled by dropping the corresponding rows from the dataset, ensuring a complete and robust dataset for analysis.

2. Exploratory Data Analysis (EDA) and Variable Selection:

- A correlation analysis was performed to identify the marketing channel most strongly correlated with Sales.
- **Key Finding:** 'TV' advertising spend exhibited the highest individual correlation with 'Sales' ($r \approx 0.999$). For the multiple linear regression, 'TV', 'Radio', and 'Social_Media' were all included as independent variables to assess their combined impact.
- A scatter plot visualizing 'TV' spend vs. 'Sales' further reinforced this strong linear trend.

3. Multiple Linear Regression Model Implementation:

- An Ordinary Least Squares (OLS) regression model was built with 'TV', 'Radio', and 'Social_Media' as independent variables and 'Sales' as the dependent variable.
- **Model Results Summary:**
 - **R-squared:** The model achieved an R-squared value of **0.999**, indicating that approximately 99.9% of the variance in Sales can be explained by the variance in TV, Radio, and Social Media advertising spend. This suggests an excellent fit of the model to the data.
 - **Coefficients:**
 - **Intercept (const):** The intercept represents the estimated sales when all advertising spends (TV, Radio, Social Media) are zero. Its statistical significance should be evaluated.
 - **TV Coefficient:** This coefficient indicates the estimated change in Sales for a one-unit increase in TV advertising spend, holding Radio and

Social Media constant.

- **Radio Coefficient:** This coefficient indicates the estimated change in Sales for a one-unit increase in Radio advertising spend, holding TV and Social Media constant.
- **Social Media Coefficient:** This coefficient indicates the estimated change in Sales for a one-unit increase in Social Media advertising spend, holding TV and Radio constant.
- **P-values:** The p-values for each coefficient indicate their statistical significance. A low p-value (typically < 0.05) suggests that the variable is a significant predictor of Sales. These values should be interpreted from the OLS summary to determine which channels have a statistically significant impact.

4. Assumption Checking:

- Diagnostic plots were generated to verify the key assumptions of linear regression:
 - **Residuals vs Fitted Plot:** This plot should show a random scatter of residuals around zero, suggesting linearity and homoscedasticity (constant variance of residuals).
 - **Normal Q-Q Plot of Residuals:** The residuals should closely follow the straight line, indicating that they are approximately normally distributed.
 - **Histogram of Residuals:** The histogram should also show a bell-shaped distribution, further supporting the normality assumption.

5. Conclusion and Business Recommendation:

- The analysis confirms a highly significant and strong positive relationship between marketing spend across the channels and Sales. The model demonstrates excellent explanatory power.
- **Actionable Recommendation:** Based on the coefficients and p-values from the multiple linear regression model, specific recommendations can be made regarding the optimal allocation of the marketing budget across TV, Radio, and Social Media to maximize ROI. Channels with statistically significant positive coefficients should be prioritized, with the magnitude of the coefficient indicating the relative impact on sales. It's crucial to interpret the p-values and confidence